Homework — 2.1.2 Thinking Ahead — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. (a) [2] 1 mark for a valid input, e.g. member ID/card, book ID/barcode, the current date. 1 mark for a valid output, e.g. the updated loan record / due-date receipt / confirmation message / updated availability of the copy. Must be data *in* vs a result *out*.

(b) [1] It is a **process** — checking for unpaid fines transforms/tests data within the routine; it is neither data flowing in nor a result flowing out.

2. [2] 1 mark precondition: the member must currently have that book **on loan** / the book's status is "issued to this member" before it can be returned. 1 mark: if unchecked, the system could mark a book returned that was never borrowed (or borrowed by someone else), corrupting loan records / inventory counts, or error on missing data.

3. (a) [2] 1 mark + 1 mark: the timetable is computed/fetched **once** and the stored result is reused for many users **[1]**, so pages load faster and the server avoids recomputing/re-querying the same data repeatedly, reducing load **[1]**.

(b) [2] 1 mark: the cached timetable can become **stale** if a train is delayed/cancelled after it was cached. 1 mark: users would then be shown **out-of-date** times unless the cache is invalidated/refreshed — so the cache needs a refresh strategy, costing extra complexity/storage.

4. [2] Any two (1 each): faster development — write once, drop into all three games; fewer bugs because the component is already **tested**; consistency — all games save/load the same way; easier maintenance — fix or improve it once and all three benefit.

5. [3] 1 mark each, applied to the order routine: **input** — e.g. items chosen, quantities, delivery address, payment details; **precondition** — e.g. the basket must be non-empty / the items must be in stock / the user logged in *before* payment is taken; **cache** — e.g. the menu/prices, or the user's saved address/payment token, stored for fast reuse. Must be applied to the scenario; generic definitions score poorly.

Part B (6 marks)

6. [2] 1 mark keep: e.g. student ID, name, year/class, the subjects/sets they take. 1 mark discard: e.g. eye colour, favourite food, home address detail — irrelevant to scheduling lessons. *(Discard need not be justified here; one keep + one discard.)*

7. [2] 1 mark: an interrupt is a signal sent to the processor that an event needs attention, causing it to pause the current task. 1 mark, any one reason: to respond promptly to I/O (key press, printer ready), errors, or hardware/timer events without constantly polling; the OS saves state, services the interrupt (via the ISR), then resumes.

8. [2] Any two (1 each): clock speed; number of cores; cache size/amount of cache; word length/bus width; pipelining. *(Accept any two valid factors.)*

Total: 14 + 6 = 20 marks